Cement Grinding Mill Operations – Reference Page 1

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 2

The Clinker Cooler must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 3

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Vertical Roller Mill	temperature	Within Design Limit	Daily
ID Fan	motor current	Normal	Monthly
Bucket Elevator	motor current	Within Design Limit	Hourly
Raw Mill Fan	feed rate	Within Design Limit	Weekly
Bucket Elevator	temperature	As per OEM Manual	Monthly

Clinker Cooler Operations – Reference Page 4

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 5

The Vertical Roller Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	pressure observed in Clinker Cooler during routine inspection.	Reported to control room engineer
02:00	power consumption observed in Clinker Cooler during routine inspection.	Within normal range
04:00	air flow observed in Limestone Crusher during routine inspection.	Reported to control room engineer
06:00	power consumption observed in Clinker Cooler during routine inspection.	Reported to control room engineer
08:00	temperature observed in Rotary Kiln during routine inspection.	Reported to control room engineer
10:00	vibration observed in Packing Machine during routine	Maintenance notified

inspection.

Packing Plant Operations – Reference Page 6

The Electrostatic Precipitator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	feed rate	Within Design Limit	Hourly
Bucket Elevator	vibration	Within Design Limit	Hourly
Rotary Kiln	pressure	Within Design Limit	Monthly
Vertical Roller Mill	temperature	Normal	Monthly
Packing Machine	pressure	Within Design Limit	Weekly

Packing Plant Operations – Reference Page 7

The Clinker Cooler must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 8

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Environmental Monitoring Procedures – Reference Page 9

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Rotary Kiln	power consumption	Normal	Daily
Limestone Crusher	power consumption	As per OEM Manual	Monthly
Vertical Roller Mill	vibration	As per OEM Manual	Hourly
Cement Separator	bearing temperature	As per OEM Manual	Hourly
Ball Mill	air flow	Within Design Limit	Daily

Limestone Quarry Operations Manual – Reference Page 10

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	feed rate observed in Bag Filter during routine inspection.	Reported to control room engineer
02:00	bearing temperature observed in Raw Mill Fan during routine inspection.	Reported to control room engineer
04:00	air flow observed in Bucket Elevator during routine inspection.	Reported to control room engineer
06:00	bearing temperature observed in Packing Machine during routine inspection.	Maintenance notified
08:00	motor current observed in Ball Mill during routine inspection.	Adjusted process parameter
10:00	power consumption observed in Bag Filter during	Adjusted process parameter

routine inspection.

Equipment Maintenance Sheets – Reference Page 11

The Vertical Roller Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 12

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	temperature	As per OEM Manual	Daily
Raw Mill Fan	power consumption	As per OEM Manual	Daily
Bucket Elevator	motor current	Normal	Daily
Ball Mill	air flow	As per OEM Manual	Weekly
Ball Mill	power consumption	Normal	Weekly

Shift Handover and Operations Logs – Reference Page 13

The Vertical Roller Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Handover and Operations Logs – Reference Page 14

The Bag Filter must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 15

The Rotary Kiln must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bag Filter	power consumption	As per OEM Manual	Weekly
Vertical Roller Mill	feed rate	Normal	Hourly
Clinker Cooler	feed rate	Normal	Hourly
Bucket Elevator	temperature	As per OEM Manual	Daily
Limestone Crusher	vibration	Within Design Limit	Weekly

Shift Log Example

Time	Observation	Action Taken
00:00	air flow observed in Bucket Elevator during routine inspection.	Reported to control room engineer
02:00	power consumption observed in ID Fan during routine inspection.	Within normal range
04:00	power consumption	Reported to control room

	observed in Raw Mill Fan during routine inspection.	engineer
06:00	air flow observed in ID Fan during routine inspection.	Reported to control room engineer
08:00	feed rate observed in Clinker Cooler during routine inspection.	Reported to control room engineer
10:00	pressure observed in Bag Filter during routine inspection.	Reported to control room engineer

Preventive Maintenance Manual – Reference Page 16

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Handover and Operations Logs – Reference Page 17

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Environmental Monitoring Procedures – Reference Page 18

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Electrostatic Precipitator	temperature	Normal	Weekly
Ball Mill	motor current	Within Design Limit	Hourly
Rotary Kiln	vibration	Within Design Limit	Weekly
Cement Separator	bearing temperature	Normal	Monthly
Ball Mill	power consumption	Within Design Limit	Daily

Plant Overview – Reference Page 19

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 20

The Bag Filter must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	feed rate observed in Cement Separator during routine inspection.	Adjusted process parameter
02:00	feed rate observed in Vertical Roller Mill during routine inspection.	Within normal range
04:00	pressure observed in Ball Mill during routine inspection.	Maintenance notified
06:00	vibration observed in Clinker Cooler during routine inspection.	Reported to control room engineer
08:00	power consumption observed in Rotary Kiln during routine inspection.	Adjusted process parameter
10:00	air flow observed in Vertical Roller Mill during routine inspection.	Maintenance notified

Troubleshooting Guide – Reference Page 21

The Limestone Crusher must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Raw Mill Fan	vibration	Within Design Limit	Daily
Packing Machine	pressure	Within Design Limit	Monthly
Clinker Cooler	power consumption	Within Design Limit	Monthly
Bag Filter	air flow	Within Design Limit	Monthly
Raw Mill Fan	vibration	Normal	Daily

Troubleshooting Guide – Reference Page 22

The Vertical Roller Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 23

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 24

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	feed rate	As per OEM Manual	Hourly
Bag Filter	feed rate	Within Design Limit	Monthly
Rotary Kiln	vibration	As per OEM Manual	Monthly
Vertical Roller Mill	air flow	Normal	Monthly
Electrostatic Precipitator	feed rate	Within Design Limit	Daily

Raw Mill Operations (VRM) – Reference Page 25

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Vertical Roller Mill during routine inspection.	Reported to control room engineer
02:00	bearing temperature observed in Clinker Cooler during routine inspection.	Within normal range
04:00	feed rate observed in Clinker Cooler during routine inspection.	Adjusted process parameter
06:00	feed rate observed in Limestone Crusher during routine inspection.	Maintenance notified
08:00	vibration observed in Clinker Cooler during routine inspection.	Maintenance notified
10:00	air flow observed in Bucket Elevator during routine	Within normal range

inspection.

Equipment Maintenance Sheets – Reference Page 26

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Clinker Cooler Operations – Reference Page 27

The Cement Separator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Cement Separator	feed rate	Within Design Limit	Monthly
Bag Filter	temperature	Normal	Hourly
Cement Separator	feed rate	As per OEM Manual	Hourly
Electrostatic Precipitator	temperature	Normal	Monthly
Clinker Cooler	vibration	Within Design Limit	Monthly

Quality Control Laboratory SOP – Reference Page 28

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 29

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 30

The Limestone Crusher must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Electrostatic Precipitator	motor current	As per OEM Manual	Weekly
Electrostatic Precipitator	temperature	Within Design Limit	Daily
Packing Machine	motor current	As per OEM Manual	Monthly
Electrostatic Precipitator	pressure	As per OEM Manual	Daily
ID Fan	pressure	Normal	Monthly

Shift Log Example

Time	Observation	Action Taken
00:00	bearing temperature observed in Electrostatic Precipitator during routine inspection.	Adjusted process parameter

02:00	motor current observed in Vertical Roller Mill during routine inspection.	Within normal range
04:00	feed rate observed in Packing Machine during routine inspection.	Reported to control room engineer
06:00	temperature observed in Ball Mill during routine inspection.	Adjusted process parameter
08:00	vibration observed in Clinker Cooler during routine inspection.	Reported to control room engineer
10:00	air flow observed in Packing Machine during routine inspection.	Within normal range

Packing Plant Operations – Reference Page 31

The Cement Separator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 32

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 33

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Rotary Kiln	bearing temperature	Normal	Hourly
Rotary Kiln	temperature	Within Design Limit	Hourly
Raw Mill Fan	pressure	As per OEM Manual	Weekly
Clinker Cooler	vibration	Normal	Monthly
Raw Mill Fan	feed rate	Normal	Weekly

Safety and Permit Procedures – Reference Page 34

The Electrostatic Precipitator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 35

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	power consumption observed in ID Fan during routine inspection.	Maintenance notified
02:00	temperature observed in Bag Filter during routine inspection.	Reported to control room engineer
04:00	pressure observed in Vertical Roller Mill during routine inspection.	Maintenance notified
06:00	temperature observed in Packing Machine during routine inspection.	Adjusted process parameter
08:00	motor current observed in Limestone Crusher during routine inspection.	Reported to control room engineer
10:00	motor current observed in Rotary Kiln during routine	Maintenance notified

inspection.

Troubleshooting Guide – Reference Page 36

The Vertical Roller Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	motor current	Within Design Limit	Daily
Limestone Crusher	bearing temperature	Normal	Monthly
Electrostatic Precipitator	pressure	Normal	Weekly
Vertical Roller Mill	feed rate	Normal	Weekly
Rotary Kiln	motor current	As per OEM Manual	Monthly

Environmental Monitoring Procedures – Reference Page 37

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Cement Grinding Mill Operations – Reference Page 38

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 39

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Rotary Kiln	pressure	Within Design Limit	Monthly
Packing Machine	power consumption	Normal	Daily
Limestone Crusher	temperature	Within Design Limit	Daily
Raw Mill Fan	motor current	As per OEM Manual	Monthly
Ball Mill	power consumption	As per OEM Manual	Hourly

Preventive Maintenance Manual – Reference Page 40

The Clinker Cooler must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Raw Mill Fan during routine inspection.	Within normal range
02:00	air flow observed in ID Fan during routine inspection.	Within normal range
04:00	temperature observed in Cement Separator during routine inspection.	Adjusted process parameter
06:00	power consumption observed in Bag Filter during routine inspection.	Adjusted process parameter
08:00	power consumption observed in Electrostatic Precipitator during routine inspection.	Reported to control room engineer
10:00	feed rate observed in Packing Machine during routine	Adjusted process parameter

inspection.

Cement Grinding Mill Operations – Reference Page 41

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 42

The Clinker Cooler must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	feed rate	Normal	Daily
Raw Mill Fan	air flow	Within Design Limit	Monthly
Bucket Elevator	pressure	Within Design Limit	Weekly
Vertical Roller Mill	bearing temperature	Within Design Limit	Hourly
Bucket Elevator	air flow	As per OEM Manual	Weekly

Plant Overview – Reference Page 43

The Bucket Elevator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 44

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 45

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	bearing temperature	Within Design Limit	Weekly
Ball Mill	vibration	Within Design Limit	Weekly
Electrostatic Precipitator	feed rate	As per OEM Manual	Hourly
Bag Filter	vibration	Normal	Weekly
ID Fan	bearing temperature	Within Design Limit	Hourly

Shift Log Example

Time	Observation	Action Taken
00:00	bearing temperature observed in Clinker Cooler during routine inspection.	Adjusted process parameter
02:00	air flow observed in ID Fan during routine inspection.	Adjusted process parameter
04:00	power consumption	Reported to control room

	observed in Raw Mill Fan during routine inspection.	engineer
06:00	feed rate observed in Cement Separator during routine inspection.	Adjusted process parameter
08:00	air flow observed in Packing Machine during routine inspection.	Maintenance notified
10:00	air flow observed in Packing Machine during routine inspection.	Maintenance notified

Limestone Quarry Operations Manual – Reference Page 46

The Packing Machine must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Handover and Operations Logs – Reference Page 47

The Ball Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 48

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bag Filter	temperature	As per OEM Manual	Daily
Bag Filter	bearing temperature	Within Design Limit	Weekly
Bucket Elevator	bearing temperature	Normal	Hourly
Raw Mill Fan	bearing temperature	Normal	Weekly
Electrostatic Precipitator	feed rate	As per OEM Manual	Monthly

Rotary Kiln Operations Manual – Reference Page 49

The Limestone Crusher must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preventive Maintenance Manual – Reference Page 50

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	power consumption observed in Limestone Crusher during routine inspection.	Within normal range
02:00	feed rate observed in ID Fan during routine inspection.	Reported to control room engineer
04:00	bearing temperature observed in Packing Machine during routine inspection.	Adjusted process parameter
06:00	power consumption observed in ID Fan during routine inspection.	Adjusted process parameter
08:00	temperature observed in Bucket Elevator during routine inspection.	Reported to control room engineer
10:00	temperature observed in Cement Separator during	Maintenance notified

routine inspection.

Raw Mill Operations (VRM) – Reference Page 51

The Cement Separator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	motor current	Normal	Hourly
Electrostatic Precipitator	vibration	As per OEM Manual	Monthly
Raw Mill Fan	vibration	Normal	Hourly
Clinker Cooler	vibration	As per OEM Manual	Monthly
Electrostatic Precipitator	motor current	Within Design Limit	Daily

Rotary Kiln Operations Manual – Reference Page 52

The ID Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 53

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 54

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	vibration	As per OEM Manual	Monthly
Ball Mill	feed rate	As per OEM Manual	Weekly
Bag Filter	motor current	Within Design Limit	Weekly
Bag Filter	vibration	Within Design Limit	Hourly
Raw Mill Fan	motor current	Within Design Limit	Weekly

Preheater and Calciner System – Reference Page 55

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Bucket Elevator during routine inspection.	Maintenance notified
02:00	feed rate observed in Raw Mill Fan during routine inspection.	Reported to control room engineer
04:00	power consumption observed in Rotary Kiln during routine inspection.	Reported to control room engineer
06:00	motor current observed in Limestone Crusher during routine inspection.	Maintenance notified
08:00	power consumption observed in Packing Machine during routine inspection.	Adjusted process parameter
10:00	air flow observed in Rotary Kiln during routine	Maintenance notified

inspection.

Troubleshooting Guide – Reference Page 56

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preventive Maintenance Manual – Reference Page 57

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Vertical Roller Mill	motor current	Within Design Limit	Hourly
Clinker Cooler	temperature	Within Design Limit	Weekly
Bucket Elevator	vibration	Within Design Limit	Weekly
ID Fan	motor current	Normal	Monthly
Cement Separator	motor current	Within Design Limit	Monthly

Quality Control Laboratory SOP – Reference Page 58

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Plant Overview – Reference Page 59

The ID Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 60

The Vertical Roller Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	feed rate	Within Design Limit	Weekly
Limestone Crusher	feed rate	Within Design Limit	Daily
Rotary Kiln	pressure	As per OEM Manual	Monthly
Bucket Elevator	feed rate	Within Design Limit	Hourly
Rotary Kiln	vibration	Normal	Hourly

Shift Log Example

Time	Observation	Action Taken
00:00	power consumption observed in Clinker Cooler during routine inspection.	Adjusted process parameter
02:00	bearing temperature observed in ID Fan during routine inspection.	Reported to control room engineer
04:00	pressure observed in ID Fan	Adjusted process parameter

	during routine inspection.	
06:00	power consumption observed in ID Fan during routine inspection.	Within normal range
08:00	motor current observed in ID Fan during routine inspection.	Adjusted process parameter
10:00	vibration observed in Cement Separator during routine inspection.	Maintenance notified

Quality Control Laboratory SOP – Reference Page 61

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 62

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 63

The Vertical Roller Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	air flow	Normal	Weekly
Rotary Kiln	feed rate	Within Design Limit	Weekly
Rotary Kiln	vibration	Within Design Limit	Weekly
Bucket Elevator	pressure	Within Design Limit	Weekly
Bucket Elevator	feed rate	As per OEM Manual	Daily

Clinker Cooler Operations – Reference Page 64

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 65

The Rotary Kiln must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	motor current observed in Bag Filter during routine inspection.	Reported to control room engineer
02:00	vibration observed in Ball Mill during routine inspection.	Within normal range
04:00	air flow observed in Rotary Kiln during routine inspection.	Maintenance notified
06:00	temperature observed in Ball Mill during routine inspection.	Reported to control room engineer
08:00	temperature observed in Electrostatic Precipitator during routine inspection.	Adjusted process parameter
10:00	air flow observed in Clinker Cooler during routine inspection.	Maintenance notified

Rotary Kiln Operations Manual – Reference Page 66

The Clinker Cooler must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Vertical Roller Mill	temperature	As per OEM Manual	Hourly
Clinker Cooler	motor current	As per OEM Manual	Monthly
Vertical Roller Mill	power consumption	Within Design Limit	Daily
Ball Mill	motor current	Within Design Limit	Hourly
Packing Machine	power consumption	As per OEM Manual	Hourly

Safety and Permit Procedures – Reference Page 67

The Ball Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 68

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 69

The Raw Mill Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	temperature	As per OEM Manual	Hourly
Rotary Kiln	motor current	Within Design Limit	Monthly
ID Fan	temperature	Within Design Limit	Hourly
Cement Separator	bearing temperature	Normal	Hourly
Packing Machine	vibration	Normal	Hourly

Clinker Cooler Operations – Reference Page 70

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	feed rate observed in Bag Filter during routine inspection.	Within normal range
02:00	motor current observed in Electrostatic Precipitator during routine inspection.	Reported to control room engineer
04:00	pressure observed in Bag Filter during routine inspection.	Reported to control room engineer
06:00	motor current observed in Electrostatic Precipitator during routine inspection.	Within normal range
08:00	motor current observed in ID Fan during routine inspection.	Adjusted process parameter
10:00	bearing temperature observed in Rotary Kiln	Maintenance notified

during routine inspection.

Cement Grinding Mill Operations – Reference Page 71

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 72

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	temperature	Normal	Daily
Vertical Roller Mill	air flow	Normal	Weekly
Ball Mill	bearing temperature	Normal	Weekly
Bag Filter	pressure	Within Design Limit	Daily
Cement Separator	vibration	Normal	Monthly

Preventive Maintenance Manual – Reference Page 73

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 74

The Packing Machine must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 75

The Cement Separator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Raw Mill Fan	temperature	As per OEM Manual	Weekly
Ball Mill	air flow	Within Design Limit	Daily
Ball Mill	vibration	As per OEM Manual	Hourly
Vertical Roller Mill	motor current	Within Design Limit	Daily
Limestone Crusher	feed rate	Within Design Limit	Hourly

Shift Log Example

Time	Observation	Action Taken
00:00	air flow observed in Clinker Cooler during routine inspection.	Reported to control room engineer
02:00	air flow observed in Ball Mill during routine inspection.	Within normal range
04:00	air flow observed in Clinker Cooler during routine	Within normal range

	inspection.	
06:00	bearing temperature observed in Ball Mill during routine inspection.	Adjusted process parameter
08:00	pressure observed in Bag Filter during routine inspection.	Reported to control room engineer
10:00	feed rate observed in ID Fan during routine inspection.	Maintenance notified

Safety and Permit Procedures – Reference Page 76

The Ball Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 77

The Clinker Cooler must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 78

The Electrostatic Precipitator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	power consumption	Within Design Limit	Weekly
Vertical Roller Mill	air flow	Normal	Hourly
Rotary Kiln	power consumption	Within Design Limit	Hourly
Bucket Elevator	temperature	Within Design Limit	Daily
Vertical Roller Mill	feed rate	Within Design Limit	Weekly

Troubleshooting Guide – Reference Page 79

The Vertical Roller Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 80

The Bucket Elevator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	feed rate observed in Packing Machine during routine inspection.	Maintenance notified
02:00	bearing temperature observed in ID Fan during routine inspection.	Adjusted process parameter
04:00	power consumption observed in Rotary Kiln during routine inspection.	Maintenance notified
06:00	pressure observed in Electrostatic Precipitator during routine inspection.	Adjusted process parameter
08:00	bearing temperature observed in Clinker Cooler during routine inspection.	Within normal range
10:00	feed rate observed in Rotary Kiln during routine	Within normal range

inspection.

Limestone Quarry Operations Manual – Reference Page 81

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	feed rate	Normal	Weekly
Clinker Cooler	power consumption	Within Design Limit	Weekly
Raw Mill Fan	pressure	As per OEM Manual	Hourly
Ball Mill	bearing temperature	As per OEM Manual	Weekly
Packing Machine	vibration	As per OEM Manual	Hourly

Rotary Kiln Operations Manual – Reference Page 82

The Clinker Cooler must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 83

The Clinker Cooler must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Handover and Operations Logs – Reference Page 84

The Cement Separator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Rotary Kiln	pressure	Within Design Limit	Weekly
Cement Separator	vibration	Within Design Limit	Monthly
Ball Mill	power consumption	As per OEM Manual	Weekly
Electrostatic Precipitator	motor current	Within Design Limit	Weekly
Vertical Roller Mill	vibration	Within Design Limit	Weekly

Preventive Maintenance Manual – Reference Page 85

The Vertical Roller Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Packing Machine during routine inspection.	Within normal range
02:00	motor current observed in Cement Separator during routine inspection.	Within normal range
04:00	bearing temperature observed in Limestone Crusher during routine inspection.	Maintenance notified
06:00	feed rate observed in Raw Mill Fan during routine inspection.	Within normal range
08:00	vibration observed in Electrostatic Precipitator during routine inspection.	Adjusted process parameter
10:00	air flow observed in Clinker	Maintenance notified

Cooler during routine inspection.

Packing Plant Operations – Reference Page 86

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 87

The Vertical Roller Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Rotary Kiln	vibration	Within Design Limit	Daily
Clinker Cooler	vibration	As per OEM Manual	Hourly
Bucket Elevator	temperature	As per OEM Manual	Daily
Rotary Kiln	pressure	Normal	Hourly
Rotary Kiln	temperature	As per OEM Manual	Hourly

Rotary Kiln Operations Manual – Reference Page 88

The Bag Filter must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 89

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Cement Grinding Mill Operations – Reference Page 90

The Raw Mill Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	feed rate	As per OEM Manual	Weekly
Packing Machine	bearing temperature	Within Design Limit	Monthly
Bag Filter	air flow	Normal	Daily
Cement Separator	bearing temperature	As per OEM Manual	Hourly
ID Fan	temperature	Within Design Limit	Daily

Shift Log Example

Time	Observation	Action Taken
00:00	air flow observed in Vertical Roller Mill during routine inspection.	Maintenance notified
02:00	feed rate observed in Packing Machine during routine inspection.	Within normal range
04:00	motor current observed in ID	Reported to control room

	Fan during routine inspection.	engineer
06:00	temperature observed in ID Fan during routine inspection.	Within normal range
08:00	feed rate observed in ID Fan during routine inspection.	Maintenance notified
10:00	power consumption observed in Packing Machine during routine inspection.	Within normal range

Rotary Kiln Operations Manual – Reference Page 91

The Rotary Kiln must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Plant Overview – Reference Page 92

The Clinker Cooler must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 93

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	temperature	Within Design Limit	Daily
Ball Mill	power consumption	As per OEM Manual	Monthly
Limestone Crusher	air flow	As per OEM Manual	Hourly
Bucket Elevator	vibration	As per OEM Manual	Weekly
Raw Mill Fan	air flow	Within Design Limit	Daily

Plant Overview – Reference Page 94

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preventive Maintenance Manual – Reference Page 95

The Bag Filter must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Rotary Kiln during routine inspection.	Maintenance notified
02:00	power consumption observed in Ball Mill during routine inspection.	Reported to control room engineer
04:00	feed rate observed in Raw Mill Fan during routine inspection.	Maintenance notified
06:00	pressure observed in Packing Machine during routine inspection.	Adjusted process parameter
08:00	motor current observed in Packing Machine during routine inspection.	Adjusted process parameter
10:00	air flow observed in Limestone Crusher during	Within normal range

routine inspection.

Packing Plant Operations – Reference Page 96

The Ball Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bag Filter	feed rate	Within Design Limit	Weekly
Ball Mill	bearing temperature	Normal	Monthly
Raw Mill Fan	pressure	As per OEM Manual	Hourly
Vertical Roller Mill	motor current	Within Design Limit	Weekly
Limestone Crusher	power consumption	Normal	Daily

Limestone Quarry Operations Manual – Reference Page 97

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 98

The Ball Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 99

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Electrostatic Precipitator	air flow	Normal	Weekly
Packing Machine	temperature	Within Design Limit	Weekly
Cement Separator	temperature	Within Design Limit	Monthly
Bucket Elevator	motor current	As per OEM Manual	Daily
Cement Separator	air flow	As per OEM Manual	Weekly

Preventive Maintenance Manual – Reference Page 100

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	air flow observed in Electrostatic Precipitator during routine inspection.	Maintenance notified
02:00	air flow observed in Electrostatic Precipitator during routine inspection.	Adjusted process parameter
04:00	feed rate observed in Raw Mill Fan during routine inspection.	Maintenance notified
06:00	power consumption observed in Electrostatic Precipitator during routine inspection.	Within normal range
08:00	feed rate observed in Rotary Kiln during routine inspection.	Adjusted process parameter
10:00	motor current observed in	Reported to control room

Clinker Cooler during routine engineer inspection.

Packing Plant Operations – Reference Page 101

The Vertical Roller Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 102

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	vibration	Within Design Limit	Monthly
Electrostatic Precipitator	motor current	As per OEM Manual	Weekly
Packing Machine	temperature	As per OEM Manual	Weekly
Rotary Kiln	motor current	As per OEM Manual	Monthly
Raw Mill Fan	air flow	As per OEM Manual	Daily

Quality Control Laboratory SOP – Reference Page 103

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 104

The Vertical Roller Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 105

The Bag Filter must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	power consumption	Within Design Limit	Monthly
ID Fan	bearing temperature	Normal	Daily
Electrostatic Precipitator	air flow	Within Design Limit	Weekly
Electrostatic Precipitator	motor current	Normal	Daily
ID Fan	pressure	Normal	Monthly

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Vertical Roller Mill during routine inspection.	Reported to control room engineer
02:00	motor current observed in Cement Separator during	Within normal range

	routine inspection.	
04:00	bearing temperature observed in Cement Separator during routine inspection.	Adjusted process parameter
06:00	air flow observed in Raw Mill Fan during routine inspection.	Adjusted process parameter
08:00	vibration observed in Clinker Cooler during routine inspection.	Maintenance notified
10:00	pressure observed in Raw Mill Fan during routine inspection.	Maintenance notified

Limestone Quarry Operations Manual – Reference Page 106

The Vertical Roller Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 107

The Vertical Roller Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 108

The Cement Separator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	pressure	Normal	Daily
Bucket Elevator	feed rate	Normal	Daily
Clinker Cooler	power consumption	Within Design Limit	Daily
Bucket Elevator	power consumption	Normal	Weekly
Vertical Roller Mill	air flow	Normal	Daily

Packing Plant Operations – Reference Page 109

The Electrostatic Precipitator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 110

The Raw Mill Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	bearing temperature observed in ID Fan during routine inspection.	Maintenance notified
02:00	motor current observed in Packing Machine during routine inspection.	Reported to control room engineer
04:00	bearing temperature observed in ID Fan during routine inspection.	Maintenance notified
06:00	air flow observed in Bag Filter during routine inspection.	Adjusted process parameter
08:00	bearing temperature observed in Clinker Cooler during routine inspection.	Within normal range
10:00	bearing temperature observed in Vertical Roller Mill during routine	Maintenance notified

inspection.

Limestone Quarry Operations Manual – Reference Page 111

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Electrostatic Precipitator	power consumption	Within Design Limit	Hourly
Raw Mill Fan	bearing temperature	Normal	Weekly
ID Fan	power consumption	Within Design Limit	Weekly
Rotary Kiln	feed rate	Normal	Weekly
ID Fan	vibration	As per OEM Manual	Weekly

Shift Handover and Operations Logs – Reference Page 112

The Ball Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 113

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 114

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bag Filter	bearing temperature	As per OEM Manual	Daily
Rotary Kiln	vibration	Normal	Daily
Limestone Crusher	feed rate	Normal	Monthly
Clinker Cooler	motor current	As per OEM Manual	Hourly
Cement Separator	motor current	Normal	Weekly

Packing Plant Operations – Reference Page 115

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	pressure observed in Ball Mill during routine inspection.	Reported to control room engineer
02:00	pressure observed in Ball Mill during routine inspection.	Adjusted process parameter
04:00	pressure observed in Bag Filter during routine inspection.	Maintenance notified
06:00	motor current observed in Ball Mill during routine inspection.	Within normal range
08:00	vibration observed in Raw Mill Fan during routine inspection.	Within normal range
10:00	bearing temperature observed in Bag Filter during routine inspection.	Within normal range

Packing Plant Operations – Reference Page 116

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 117

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Cement Separator	motor current	Within Design Limit	Hourly
Raw Mill Fan	motor current	Within Design Limit	Hourly
Ball Mill	vibration	As per OEM Manual	Weekly
Bag Filter	motor current	As per OEM Manual	Daily
Vertical Roller Mill	pressure	Normal	Hourly

Shift Handover and Operations Logs – Reference Page 118

The Ball Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 119

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Handover and Operations Logs – Reference Page 120

The Ball Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Cement Separator	motor current	Within Design Limit	Monthly
ID Fan	vibration	Within Design Limit	Monthly
Limestone Crusher	power consumption	Within Design Limit	Hourly
Vertical Roller Mill	feed rate	As per OEM Manual	Monthly
Clinker Cooler	power consumption	As per OEM Manual	Daily

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Vertical Roller Mill during routine inspection.	Adjusted process parameter
02:00	air flow observed in Clinker Cooler during routine inspection.	Reported to control room engineer
04:00	bearing temperature	Adjusted process parameter

	observed in Bucket Elevator during routine inspection.	
06:00	temperature observed in Electrostatic Precipitator during routine inspection.	Within normal range
08:00	motor current observed in Rotary Kiln during routine inspection.	Within normal range
10:00	motor current observed in Bucket Elevator during routine inspection.	Maintenance notified

Troubleshooting Guide – Reference Page 121

The Ball Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 122

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 123

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	air flow	Normal	Hourly
ID Fan	feed rate	Within Design Limit	Daily
ID Fan	power consumption	Normal	Monthly
Ball Mill	power consumption	Within Design Limit	Hourly
Packing Machine	feed rate	Normal	Hourly

Packing Plant Operations – Reference Page 124

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preventive Maintenance Manual – Reference Page 125

The Packing Machine must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Electrostatic Precipitator during routine inspection.	Maintenance notified
02:00	power consumption observed in Vertical Roller Mill during routine inspection.	Within normal range
04:00	pressure observed in Limestone Crusher during routine inspection.	Reported to control room engineer
06:00	feed rate observed in Electrostatic Precipitator during routine inspection.	Within normal range
08:00	motor current observed in Clinker Cooler during routine inspection.	Adjusted process parameter
10:00	vibration observed in Ball Mill	Reported to control room

during routine inspection. engineer

Clinker Cooler Operations – Reference Page 126

The Packing Machine must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Raw Mill Fan	air flow	Normal	Hourly
ID Fan	temperature	Normal	Daily
Vertical Roller Mill	power consumption	Within Design Limit	Weekly
Rotary Kiln	power consumption	Within Design Limit	Monthly
Bucket Elevator	pressure	Normal	Monthly

Crusher House Operations – Reference Page 127

The Packing Machine must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 128

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Troubleshooting Guide – Reference Page 129

The Ball Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Vertical Roller Mill	power consumption	Normal	Weekly
Rotary Kiln	bearing temperature	Within Design Limit	Hourly
Rotary Kiln	pressure	Normal	Hourly
ID Fan	power consumption	Within Design Limit	Monthly
Electrostatic Precipitator	power consumption	Within Design Limit	Daily

Equipment Maintenance Sheets – Reference Page 130

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Ball Mill during routine inspection.	Reported to control room engineer
02:00	temperature observed in Electrostatic Precipitator during routine inspection.	Reported to control room engineer
04:00	vibration observed in Vertical Roller Mill during routine inspection.	Adjusted process parameter
06:00	temperature observed in Rotary Kiln during routine inspection.	Adjusted process parameter
08:00	bearing temperature observed in Vertical Roller Mill during routine inspection.	Adjusted process parameter
10:00	feed rate observed in Clinker Cooler during routine	Adjusted process parameter

inspection.

Preventive Maintenance Manual – Reference Page 131

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Environmental Monitoring Procedures – Reference Page 132

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
ID Fan	vibration	Within Design Limit	Daily
ID Fan	vibration	Within Design Limit	Monthly
ID Fan	motor current	Within Design Limit	Hourly
Ball Mill	vibration	Normal	Hourly
Limestone Crusher	vibration	Within Design Limit	Daily

Crusher House Operations – Reference Page 133

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 134

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Plant Overview – Reference Page 135

The Packing Machine must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Raw Mill Fan	motor current	As per OEM Manual	Hourly
Packing Machine	power consumption	As per OEM Manual	Monthly
Electrostatic Precipitator	bearing temperature	Normal	Hourly
Bag Filter	temperature	Within Design Limit	Monthly
Ball Mill	feed rate	As per OEM Manual	Monthly

Shift Log Example

Time	Observation	Action Taken
00:00	feed rate observed in Bucket Elevator during routine inspection.	Reported to control room engineer
02:00	feed rate observed in Packing Machine during routine inspection.	Maintenance notified

04:00	temperature observed in Vertical Roller Mill during routine inspection.	Reported to control room engineer
06:00	feed rate observed in Electrostatic Precipitator during routine inspection.	Adjusted process parameter
08:00	temperature observed in Raw Mill Fan during routine inspection.	Maintenance notified
10:00	bearing temperature observed in Raw Mill Fan during routine inspection.	Adjusted process parameter

Rotary Kiln Operations Manual – Reference Page 136

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Cement Grinding Mill Operations – Reference Page 137

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 138

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Cement Separator	bearing temperature	Normal	Monthly
Electrostatic Precipitator	motor current	Normal	Weekly
Bag Filter	feed rate	Normal	Weekly
Vertical Roller Mill	power consumption	Within Design Limit	Weekly
Clinker Cooler	bearing temperature	Normal	Daily

Clinker Cooler Operations – Reference Page 139

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 140

The Raw Mill Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	feed rate observed in Bucket Elevator during routine inspection.	Maintenance notified
02:00	air flow observed in Rotary Kiln during routine inspection.	Adjusted process parameter
04:00	pressure observed in Bucket Elevator during routine inspection.	Within normal range
06:00	vibration observed in Clinker Cooler during routine inspection.	Maintenance notified
08:00	air flow observed in Clinker Cooler during routine inspection.	Within normal range
10:00	temperature observed in Clinker Cooler during routine	Adjusted process parameter

inspection.

Troubleshooting Guide – Reference Page 141

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	temperature	Normal	Daily
Bucket Elevator	temperature	Within Design Limit	Daily
ID Fan	motor current	As per OEM Manual	Weekly
Packing Machine	temperature	As per OEM Manual	Hourly
Clinker Cooler	bearing temperature	Normal	Monthly

Packing Plant Operations – Reference Page 142

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preventive Maintenance Manual – Reference Page 143

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Clinker Cooler Operations – Reference Page 144

The Limestone Crusher must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Packing Machine	power consumption	As per OEM Manual	Weekly
Cement Separator	air flow	Within Design Limit	Daily
Vertical Roller Mill	air flow	Normal	Daily
Bucket Elevator	motor current	Normal	Hourly
Cement Separator	power consumption	Within Design Limit	Monthly

Packing Plant Operations – Reference Page 145

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Vertical Roller Mill during routine inspection.	Adjusted process parameter
02:00	air flow observed in Raw Mill Fan during routine inspection.	Within normal range
04:00	temperature observed in Electrostatic Precipitator during routine inspection.	Maintenance notified
06:00	vibration observed in Bucket Elevator during routine inspection.	Adjusted process parameter
08:00	motor current observed in Rotary Kiln during routine inspection.	Maintenance notified
10:00	temperature observed in Raw Mill Fan during routine	Within normal range

inspection.

Limestone Quarry Operations Manual – Reference Page 146

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Environmental Monitoring Procedures – Reference Page 147

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	power consumption	Normal	Daily
Bag Filter	pressure	Normal	Daily
Packing Machine	vibration	Within Design Limit	Monthly
Vertical Roller Mill	vibration	Normal	Daily
Limestone Crusher	feed rate	Normal	Weekly

Safety and Permit Procedures – Reference Page 148

The Bag Filter must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 149

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 150

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Raw Mill Fan	temperature	Within Design Limit	Hourly
Ball Mill	bearing temperature	Normal	Weekly
Clinker Cooler	temperature	As per OEM Manual	Weekly
Cement Separator	feed rate	Normal	Daily
Limestone Crusher	feed rate	Normal	Hourly

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Bag Filter during routine inspection.	Adjusted process parameter
02:00	bearing temperature observed in Packing Machine during routine inspection.	Maintenance notified
04:00	motor current observed in	Reported to control room

	Ball Mill during routine inspection.	engineer
06:00	pressure observed in Limestone Crusher during routine inspection.	Adjusted process parameter
08:00	bearing temperature observed in Bucket Elevator during routine inspection.	Adjusted process parameter
10:00	power consumption observed in Limestone Crusher during routine inspection.	Adjusted process parameter

Packing Plant Operations – Reference Page 151

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 152

The Cement Separator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 153

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Raw Mill Fan	air flow	As per OEM Manual	Weekly
Vertical Roller Mill	temperature	Within Design Limit	Weekly
Cement Separator	pressure	As per OEM Manual	Weekly
Limestone Crusher	motor current	Within Design Limit	Weekly
ID Fan	air flow	Within Design Limit	Daily

Environmental Monitoring Procedures – Reference Page 154

The ID Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 155

The Ball Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Electrostatic Precipitator during routine inspection.	Within normal range
02:00	air flow observed in Ball Mill during routine inspection.	Maintenance notified
04:00	air flow observed in Raw Mill Fan during routine inspection.	Within normal range
06:00	pressure observed in Raw Mill Fan during routine inspection.	Adjusted process parameter
08:00	motor current observed in Bucket Elevator during routine inspection.	Reported to control room engineer
10:00	air flow observed in Ball Mill during routine inspection.	Reported to control room engineer

Plant Overview – Reference Page 156

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	air flow	As per OEM Manual	Weekly
ID Fan	air flow	As per OEM Manual	Daily
Bucket Elevator	air flow	As per OEM Manual	Monthly
Rotary Kiln	bearing temperature	Normal	Monthly
Vertical Roller Mill	motor current	Normal	Monthly

Rotary Kiln Operations Manual – Reference Page 157

The Electrostatic Precipitator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 158

The Clinker Cooler must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 159

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	power consumption	Within Design Limit	Weekly
Packing Machine	temperature	Within Design Limit	Hourly
Vertical Roller Mill	motor current	Within Design Limit	Daily
Ball Mill	vibration	Normal	Daily
Limestone Crusher	vibration	Within Design Limit	Weekly

Preventive Maintenance Manual – Reference Page 160

The Electrostatic Precipitator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	power consumption observed in Vertical Roller Mill during routine inspection.	Adjusted process parameter
02:00	bearing temperature observed in Packing Machine during routine inspection.	Maintenance notified
04:00	feed rate observed in Electrostatic Precipitator during routine inspection.	Within normal range
06:00	power consumption observed in Bucket Elevator during routine inspection.	Reported to control room engineer
08:00	vibration observed in Clinker Cooler during routine inspection.	Reported to control room engineer
10:00	bearing temperature	Maintenance notified

observed in Rotary Kiln
during routine inspection.

Preheater and Calciner System – Reference Page 161

The Raw Mill Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preventive Maintenance Manual – Reference Page 162

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Vertical Roller Mill	temperature	Normal	Weekly
Bag Filter	feed rate	Within Design Limit	Hourly
Raw Mill Fan	feed rate	Within Design Limit	Weekly
Vertical Roller Mill	air flow	Normal	Hourly
Rotary Kiln	air flow	Within Design Limit	Weekly

Rotary Kiln Operations Manual – Reference Page 163

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 164

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Cement Grinding Mill Operations – Reference Page 165

The Raw Mill Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Rotary Kiln	feed rate	Normal	Monthly
ID Fan	feed rate	Within Design Limit	Hourly
Electrostatic Precipitator	air flow	Within Design Limit	Weekly
Limestone Crusher	motor current	Within Design Limit	Daily
Limestone Crusher	air flow	As per OEM Manual	Daily

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Clinker Cooler during routine inspection.	Maintenance notified
02:00	air flow observed in Rotary Kiln during routine inspection.	Reported to control room engineer

04:00	air flow observed in Rotary Kiln during routine inspection.	Adjusted process parameter
06:00	power consumption observed in Bucket Elevator during routine inspection.	Adjusted process parameter
08:00	air flow observed in Cement Separator during routine inspection.	Adjusted process parameter
10:00	power consumption observed in Vertical Roller Mill during routine inspection.	Maintenance notified

Rotary Kiln Operations Manual – Reference Page 166

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Handover and Operations Logs – Reference Page 167

The Bucket Elevator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Handover and Operations Logs – Reference Page 168

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Packing Machine	bearing temperature	Within Design Limit	Daily
Electrostatic Precipitator	temperature	As per OEM Manual	Hourly
ID Fan	temperature	Within Design Limit	Daily
ID Fan	air flow	Within Design Limit	Daily
Electrostatic Precipitator	vibration	Within Design Limit	Hourly

Troubleshooting Guide – Reference Page 169

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 170

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	feed rate observed in Cement Separator during routine inspection.	Maintenance notified
02:00	temperature observed in Packing Machine during routine inspection.	Adjusted process parameter
04:00	pressure observed in Packing Machine during routine inspection.	Maintenance notified
06:00	bearing temperature observed in Cement Separator during routine inspection.	Adjusted process parameter
08:00	motor current observed in ID Fan during routine inspection.	Adjusted process parameter
10:00	bearing temperature	Within normal range

observed in Vertical Roller Mill during routine inspection.

Equipment Maintenance Sheets – Reference Page 171

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	vibration	As per OEM Manual	Monthly
Limestone Crusher	motor current	Within Design Limit	Monthly
Clinker Cooler	power consumption	Normal	Weekly
Bucket Elevator	air flow	As per OEM Manual	Hourly
Limestone Crusher	vibration	Within Design Limit	Daily

Rotary Kiln Operations Manual – Reference Page 172

The Rotary Kiln must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 173

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Troubleshooting Guide – Reference Page 174

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	bearing temperature	As per OEM Manual	Daily
Rotary Kiln	motor current	Normal	Weekly
Electrostatic Precipitator	temperature	Normal	Daily
Electrostatic Precipitator	feed rate	Normal	Weekly
ID Fan	motor current	Within Design Limit	Daily

Rotary Kiln Operations Manual – Reference Page 175

The Ball Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Electrostatic Precipitator during routine inspection.	Maintenance notified
02:00	pressure observed in Rotary Kiln during routine inspection.	Maintenance notified
04:00	air flow observed in Bag Filter during routine inspection.	Within normal range
06:00	bearing temperature observed in Ball Mill during routine inspection.	Reported to control room engineer
08:00	temperature observed in Ball Mill during routine inspection.	Reported to control room engineer
10:00	air flow observed in ID Fan during routine inspection.	Within normal range

Safety and Permit Procedures – Reference Page 176

The Electrostatic Precipitator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Environmental Monitoring Procedures – Reference Page 177

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Vertical Roller Mill	motor current	As per OEM Manual	Hourly
Cement Separator	temperature	Normal	Daily
Limestone Crusher	vibration	Normal	Monthly
Limestone Crusher	pressure	Normal	Weekly
Rotary Kiln	vibration	Normal	Monthly

Shift Handover and Operations Logs – Reference Page 178

The Vertical Roller Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 179

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 180

The Raw Mill Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
ID Fan	vibration	Within Design Limit	Hourly
Limestone Crusher	air flow	Within Design Limit	Monthly
Limestone Crusher	power consumption	As per OEM Manual	Monthly
Cement Separator	feed rate	As per OEM Manual	Hourly
Raw Mill Fan	feed rate	Normal	Daily

Shift Log Example

Time	Observation	Action Taken
00:00	motor current observed in Rotary Kiln during routine inspection.	Within normal range
02:00	power consumption observed in Bucket Elevator during routine inspection.	Reported to control room engineer
04:00	bearing temperature	Maintenance notified

	observed in Bag Filter during routine inspection.	
06:00	feed rate observed in Bag Filter during routine inspection.	Reported to control room engineer
08:00	bearing temperature observed in Rotary Kiln during routine inspection.	Adjusted process parameter
10:00	vibration observed in Bag Filter during routine inspection.	Within normal range

Quality Control Laboratory SOP – Reference Page 181

The Electrostatic Precipitator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Clinker Cooler Operations – Reference Page 182

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 183

The Rotary Kiln must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Raw Mill Fan	vibration	Normal	Weekly
Ball Mill	feed rate	Within Design Limit	Weekly
Ball Mill	power consumption	Normal	Weekly
Packing Machine	motor current	As per OEM Manual	Hourly
Packing Machine	vibration	As per OEM Manual	Daily

Preheater and Calciner System – Reference Page 184

The Raw Mill Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 185

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	pressure observed in Raw Mill Fan during routine inspection.	Reported to control room engineer
02:00	air flow observed in Ball Mill during routine inspection.	Adjusted process parameter
04:00	power consumption observed in Raw Mill Fan during routine inspection.	Adjusted process parameter
06:00	vibration observed in ID Fan during routine inspection.	Adjusted process parameter
08:00	temperature observed in Bag Filter during routine inspection.	Within normal range
10:00	bearing temperature observed in Rotary Kiln during routine inspection.	Within normal range

Cement Grinding Mill Operations – Reference Page 186

The Ball Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	bearing temperature	Normal	Daily
Rotary Kiln	power consumption	Normal	Weekly
ID Fan	power consumption	Within Design Limit	Hourly
Raw Mill Fan	pressure	As per OEM Manual	Daily
Electrostatic Precipitator	power consumption	Within Design Limit	Weekly

Safety and Permit Procedures – Reference Page 187

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Environmental Monitoring Procedures – Reference Page 188

The Raw Mill Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 189

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	temperature	Within Design Limit	Weekly
Limestone Crusher	temperature	Within Design Limit	Hourly
Limestone Crusher	motor current	Normal	Hourly
Ball Mill	pressure	Normal	Weekly
Vertical Roller Mill	power consumption	As per OEM Manual	Daily

Clinker Cooler Operations – Reference Page 190

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	motor current observed in Bucket Elevator during routine inspection.	Within normal range
02:00	motor current observed in ID Fan during routine inspection.	Maintenance notified
04:00	vibration observed in Raw Mill Fan during routine inspection.	Within normal range
06:00	vibration observed in Rotary Kiln during routine inspection.	Maintenance notified
08:00	motor current observed in Packing Machine during routine inspection.	Maintenance notified
10:00	motor current observed in Packing Machine during	Within normal range

routine inspection.

Equipment Maintenance Sheets – Reference Page 191

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 192

The Clinker Cooler must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	feed rate	Within Design Limit	Hourly
Bag Filter	temperature	Within Design Limit	Hourly
Clinker Cooler	temperature	Within Design Limit	Weekly
Rotary Kiln	feed rate	Normal	Hourly
Packing Machine	vibration	Normal	Weekly

Safety and Permit Procedures – Reference Page 193

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Plant Overview – Reference Page 194

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Clinker Cooler Operations – Reference Page 195

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	motor current	Normal	Hourly
ID Fan	pressure	As per OEM Manual	Hourly
ID Fan	power consumption	As per OEM Manual	Weekly
Bucket Elevator	bearing temperature	As per OEM Manual	Hourly
Cement Separator	motor current	Normal	Weekly

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Packing Machine during routine inspection.	Maintenance notified
02:00	air flow observed in Bag Filter during routine inspection.	Adjusted process parameter
04:00	pressure observed in Vertical Roller Mill during routine	Maintenance notified

	inspection.	
06:00	air flow observed in Bag Filter during routine inspection.	Within normal range
08:00	vibration observed in Ball Mill during routine inspection.	Reported to control room engineer
10:00	air flow observed in Packing Machine during routine inspection.	Maintenance notified

Shift Handover and Operations Logs – Reference Page 196

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Troubleshooting Guide – Reference Page 197

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Handover and Operations Logs – Reference Page 198

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	bearing temperature	Normal	Daily
ID Fan	vibration	Normal	Hourly
Limestone Crusher	power consumption	Normal	Monthly
Cement Separator	power consumption	As per OEM Manual	Hourly
ID Fan	feed rate	As per OEM Manual	Monthly

Safety and Permit Procedures – Reference Page 199

The Vertical Roller Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 200

The Vertical Roller Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	motor current observed in Vertical Roller Mill during routine inspection.	Reported to control room engineer
02:00	motor current observed in Electrostatic Precipitator during routine inspection.	Maintenance notified
04:00	feed rate observed in Cement Separator during routine inspection.	Adjusted process parameter
06:00	pressure observed in Bag Filter during routine inspection.	Maintenance notified
08:00	vibration observed in Vertical Roller Mill during routine inspection.	Maintenance notified
10:00	feed rate observed in Clinker Cooler during routine	Within normal range

inspection.

Raw Mill Operations (VRM) – Reference Page 201

The Cement Separator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	bearing temperature	Normal	Weekly
Cement Separator	temperature	As per OEM Manual	Hourly
Electrostatic Precipitator	pressure	As per OEM Manual	Weekly
Ball Mill	feed rate	Within Design Limit	Daily
Packing Machine	bearing temperature	Within Design Limit	Monthly

Limestone Quarry Operations Manual – Reference Page 202

The ID Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 203

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 204

The Vertical Roller Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Packing Machine	motor current	Within Design Limit	Daily
Clinker Cooler	power consumption	Within Design Limit	Daily
Packing Machine	motor current	Normal	Hourly
Cement Separator	power consumption	Within Design Limit	Monthly
Raw Mill Fan	motor current	As per OEM Manual	Weekly

Environmental Monitoring Procedures – Reference Page 205

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Ball Mill during routine inspection.	Adjusted process parameter
02:00	feed rate observed in ID Fan during routine inspection.	Within normal range
04:00	vibration observed in ID Fan during routine inspection.	Adjusted process parameter
06:00	bearing temperature observed in Limestone Crusher during routine inspection.	Maintenance notified
08:00	feed rate observed in Vertical Roller Mill during routine inspection.	Reported to control room engineer
10:00	vibration observed in Limestone Crusher during routine inspection.	Within normal range

Cement Grinding Mill Operations – Reference Page 206

The Rotary Kiln must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 207

The Raw Mill Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Electrostatic Precipitator	power consumption	Normal	Hourly
Vertical Roller Mill	pressure	Normal	Monthly
Clinker Cooler	temperature	Within Design Limit	Daily
ID Fan	bearing temperature	As per OEM Manual	Daily
Rotary Kiln	air flow	Within Design Limit	Hourly

Raw Mill Operations (VRM) – Reference Page 208

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Environmental Monitoring Procedures – Reference Page 209

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 210

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	temperature	Normal	Daily
Cement Separator	bearing temperature	Within Design Limit	Monthly
Rotary Kiln	power consumption	As per OEM Manual	Monthly
Limestone Crusher	pressure	Within Design Limit	Daily
Raw Mill Fan	feed rate	As per OEM Manual	Daily

Shift Log Example

Time	Observation	Action Taken
00:00	pressure observed in Bag Filter during routine inspection.	Maintenance notified
02:00	power consumption observed in Electrostatic Precipitator during routine inspection.	Maintenance notified

04:00	bearing temperature observed in Vertical Roller Mill during routine inspection.	Maintenance notified
06:00	air flow observed in Packing Machine during routine inspection.	Reported to control room engineer
08:00	bearing temperature observed in Packing Machine during routine inspection.	Maintenance notified
10:00	air flow observed in Clinker Cooler during routine inspection.	Within normal range

Preheater and Calciner System – Reference Page 211

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 212

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Troubleshooting Guide – Reference Page 213

The Raw Mill Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Rotary Kiln	motor current	Normal	Hourly
Packing Machine	temperature	Within Design Limit	Daily
Bag Filter	bearing temperature	Within Design Limit	Hourly
Cement Separator	pressure	Normal	Hourly
Electrostatic Precipitator	bearing temperature	Normal	Daily

Preventive Maintenance Manual – Reference Page 214

The Rotary Kiln must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 215

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	bearing temperature observed in Limestone Crusher during routine inspection.	Reported to control room engineer
02:00	temperature observed in Vertical Roller Mill during routine inspection.	Reported to control room engineer
04:00	power consumption observed in Vertical Roller Mill during routine inspection.	Maintenance notified
06:00	bearing temperature observed in Packing Machine during routine inspection.	Adjusted process parameter
08:00	temperature observed in Clinker Cooler during routine inspection.	Reported to control room engineer

10:00	power consumption observed in Bucket Elevator during routine inspection.	Within normal range

Preventive Maintenance Manual – Reference Page 216

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	power consumption	Within Design Limit	Daily
Cement Separator	bearing temperature	As per OEM Manual	Hourly
ID Fan	air flow	As per OEM Manual	Hourly
Limestone Crusher	air flow	As per OEM Manual	Monthly
Clinker Cooler	pressure	Normal	Monthly

Raw Mill Operations (VRM) – Reference Page 217

The Vertical Roller Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Clinker Cooler Operations – Reference Page 218

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Cement Grinding Mill Operations – Reference Page 219

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
ID Fan	feed rate	Within Design Limit	Monthly
Raw Mill Fan	power consumption	Normal	Weekly
Vertical Roller Mill	motor current	Within Design Limit	Daily
Bag Filter	power consumption	As per OEM Manual	Hourly
Bag Filter	feed rate	Within Design Limit	Monthly

Crusher House Operations – Reference Page 220

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	feed rate observed in Cement Separator during routine inspection.	Reported to control room engineer
02:00	power consumption observed in Cement Separator during routine inspection.	Adjusted process parameter
04:00	bearing temperature observed in Cement Separator during routine inspection.	Within normal range
06:00	bearing temperature observed in Raw Mill Fan during routine inspection.	Adjusted process parameter
08:00	motor current observed in Limestone Crusher during routine inspection.	Adjusted process parameter

10:00	temperature observed in Raw Mill Fan during routine inspection.	Maintenance notified

Safety and Permit Procedures – Reference Page 221

The Vertical Roller Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 222

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Electrostatic Precipitator	pressure	As per OEM Manual	Hourly
Limestone Crusher	power consumption	Within Design Limit	Daily
Limestone Crusher	temperature	Normal	Hourly
Bucket Elevator	power consumption	Within Design Limit	Daily
Cement Separator	feed rate	Normal	Weekly

Limestone Quarry Operations Manual – Reference Page 223

The Rotary Kiln must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Clinker Cooler Operations – Reference Page 224

The Vertical Roller Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 225

The Limestone Crusher must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bag Filter	bearing temperature	As per OEM Manual	Monthly
Clinker Cooler	motor current	As per OEM Manual	Weekly
Clinker Cooler	vibration	As per OEM Manual	Monthly
Ball Mill	feed rate	Within Design Limit	Daily
Packing Machine	motor current	Normal	Weekly

Shift Log Example

Time	Observation	Action Taken
00:00	power consumption observed in Electrostatic Precipitator during routine inspection.	Reported to control room engineer
02:00	bearing temperature observed in Clinker Cooler during routine inspection.	Maintenance notified

04:00	motor current observed in Clinker Cooler during routine inspection.	Reported to control room engineer
06:00	power consumption observed in Ball Mill during routine inspection.	Within normal range
08:00	feed rate observed in Clinker Cooler during routine inspection.	Within normal range
10:00	power consumption observed in Rotary Kiln during routine inspection.	Reported to control room engineer

Crusher House Operations – Reference Page 226

The Rotary Kiln must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 227

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preventive Maintenance Manual – Reference Page 228

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Raw Mill Fan	bearing temperature	Normal	Weekly
Bag Filter	motor current	As per OEM Manual	Monthly
Cement Separator	feed rate	Normal	Weekly
Ball Mill	bearing temperature	Within Design Limit	Weekly
Packing Machine	motor current	Within Design Limit	Daily

Safety and Permit Procedures – Reference Page 229

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 230

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	air flow observed in Vertical Roller Mill during routine inspection.	Reported to control room engineer
02:00	power consumption observed in Rotary Kiln during routine inspection.	Within normal range
04:00	vibration observed in Clinker Cooler during routine inspection.	Adjusted process parameter
06:00	temperature observed in Vertical Roller Mill during routine inspection.	Within normal range
08:00	temperature observed in Clinker Cooler during routine inspection.	Adjusted process parameter
10:00	power consumption observed in Bag Filter during	Within normal range

routine inspection.

Packing Plant Operations – Reference Page 231

The Vertical Roller Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	bearing temperature	Within Design Limit	Monthly
Raw Mill Fan	motor current	As per OEM Manual	Monthly
Vertical Roller Mill	motor current	Normal	Monthly
Vertical Roller Mill	vibration	Normal	Daily
Limestone Crusher	air flow	Normal	Hourly

Clinker Cooler Operations – Reference Page 232

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 233

The Ball Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 234

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Packing Machine	power consumption	As per OEM Manual	Hourly
Ball Mill	pressure	As per OEM Manual	Monthly
Cement Separator	bearing temperature	Normal	Weekly
Electrostatic Precipitator	air flow	Within Design Limit	Weekly
Rotary Kiln	temperature	As per OEM Manual	Hourly

Environmental Monitoring Procedures – Reference Page 235

The Ball Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	bearing temperature observed in Packing Machine during routine inspection.	Within normal range
02:00	temperature observed in Bag Filter during routine inspection.	Reported to control room engineer
04:00	temperature observed in Clinker Cooler during routine inspection.	Maintenance notified
06:00	feed rate observed in Ball Mill during routine inspection.	Reported to control room engineer
08:00	power consumption observed in Limestone Crusher during routine inspection.	Adjusted process parameter
10:00	temperature observed in	Maintenance notified

Rotary Kiln during routine inspection.

Cement Grinding Mill Operations – Reference Page 236

The Rotary Kiln must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 237

The ID Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	pressure	Within Design Limit	Hourly
Cement Separator	power consumption	Normal	Monthly
Vertical Roller Mill	feed rate	As per OEM Manual	Monthly
Limestone Crusher	temperature	Within Design Limit	Hourly
Bag Filter	temperature	Within Design Limit	Monthly

Shift Handover and Operations Logs – Reference Page 238

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preventive Maintenance Manual – Reference Page 239

The Vertical Roller Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 240

The Limestone Crusher must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	pressure	Normal	Monthly
Packing Machine	motor current	Within Design Limit	Daily
Vertical Roller Mill	power consumption	Normal	Monthly
Limestone Crusher	feed rate	Within Design Limit	Weekly
ID Fan	temperature	Normal	Monthly

Shift Log Example

Time	Observation	Action Taken
00:00	feed rate observed in Rotary Kiln during routine inspection.	Adjusted process parameter
02:00	air flow observed in ID Fan during routine inspection.	Adjusted process parameter
04:00	bearing temperature observed in Vertical Roller	Reported to control room engineer

	Mill during routine inspection.	
06:00	motor current observed in Limestone Crusher during routine inspection.	Within normal range
08:00	vibration observed in Bag Filter during routine inspection.	Within normal range
10:00	vibration observed in Electrostatic Precipitator during routine inspection.	Reported to control room engineer

Crusher House Operations – Reference Page 241

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 242

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Handover and Operations Logs – Reference Page 243

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Electrostatic Precipitator	bearing temperature	As per OEM Manual	Daily
Limestone Crusher	air flow	Normal	Hourly
Clinker Cooler	bearing temperature	Normal	Daily
Ball Mill	feed rate	Normal	Weekly
ID Fan	pressure	Normal	Hourly

Preventive Maintenance Manual – Reference Page 244

The Vertical Roller Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preventive Maintenance Manual – Reference Page 245

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	pressure observed in Clinker Cooler during routine inspection.	Maintenance notified
02:00	feed rate observed in Limestone Crusher during routine inspection.	Maintenance notified
04:00	air flow observed in Raw Mill Fan during routine inspection.	Reported to control room engineer
06:00	bearing temperature observed in Rotary Kiln during routine inspection.	Adjusted process parameter
08:00	feed rate observed in Packing Machine during routine inspection.	Reported to control room engineer
10:00	feed rate observed in Ball Mill during routine	Within normal range

inspection.

Limestone Quarry Operations Manual – Reference Page 246

The Clinker Cooler must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Rotary Kiln	motor current	As per OEM Manual	Hourly
Vertical Roller Mill	motor current	Normal	Hourly
Vertical Roller Mill	temperature	Within Design Limit	Weekly
Electrostatic Precipitator	feed rate	Within Design Limit	Monthly
Cement Separator	air flow	Within Design Limit	Daily

Quality Control Laboratory SOP – Reference Page 247

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Troubleshooting Guide – Reference Page 248

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 249

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	feed rate	Within Design Limit	Weekly
Vertical Roller Mill	air flow	Within Design Limit	Hourly
Bucket Elevator	feed rate	Within Design Limit	Hourly
Rotary Kiln	vibration	Within Design Limit	Monthly
Limestone Crusher	temperature	Normal	Daily

Troubleshooting Guide – Reference Page 250

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Electrostatic Precipitator during routine inspection.	Maintenance notified
02:00	air flow observed in Packing Machine during routine inspection.	Within normal range
04:00	feed rate observed in Limestone Crusher during routine inspection.	Within normal range
06:00	air flow observed in ID Fan during routine inspection.	Maintenance notified
08:00	temperature observed in Ball Mill during routine inspection.	Within normal range
10:00	feed rate observed in Vertical Roller Mill during routine inspection.	Within normal range

Crusher House Operations – Reference Page 251

The Raw Mill Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 252

The Vertical Roller Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
ID Fan	motor current	Within Design Limit	Hourly
Ball Mill	feed rate	Normal	Monthly
Cement Separator	air flow	Within Design Limit	Daily
Packing Machine	power consumption	Within Design Limit	Monthly
Ball Mill	bearing temperature	As per OEM Manual	Hourly

Preheater and Calciner System – Reference Page 253

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 254

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Environmental Monitoring Procedures – Reference Page 255

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bag Filter	pressure	Within Design Limit	Hourly
Vertical Roller Mill	vibration	Within Design Limit	Hourly
Clinker Cooler	vibration	As per OEM Manual	Monthly
Ball Mill	motor current	Within Design Limit	Weekly
ID Fan	bearing temperature	Within Design Limit	Hourly

Shift Log Example

Time	Observation	Action Taken
00:00	air flow observed in Bag Filter during routine inspection.	Within normal range
02:00	temperature observed in Bag Filter during routine inspection.	Maintenance notified
04:00	power consumption observed in Cement	Within normal range

	Separator during routine inspection.	
06:00	vibration observed in Bucket Elevator during routine inspection.	Reported to control room engineer
08:00	pressure observed in Ball Mill during routine inspection.	Adjusted process parameter
10:00	motor current observed in Vertical Roller Mill during routine inspection.	Reported to control room engineer

Plant Overview – Reference Page 256

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 257

The Raw Mill Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 258

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	vibration	Within Design Limit	Monthly
ID Fan	bearing temperature	As per OEM Manual	Monthly
Ball Mill	temperature	As per OEM Manual	Monthly
Vertical Roller Mill	air flow	As per OEM Manual	Monthly
Rotary Kiln	vibration	As per OEM Manual	Hourly

Packing Plant Operations – Reference Page 259

The Cement Separator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Clinker Cooler Operations – Reference Page 260

The Vertical Roller Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	air flow observed in Raw Mill Fan during routine inspection.	Reported to control room engineer
02:00	bearing temperature observed in Rotary Kiln during routine inspection.	Maintenance notified
04:00	feed rate observed in Ball Mill during routine inspection.	Within normal range
06:00	feed rate observed in Bag Filter during routine inspection.	Maintenance notified
08:00	power consumption observed in Rotary Kiln during routine inspection.	Within normal range
10:00	feed rate observed in Rotary Kiln during routine	Within normal range

inspection.

Plant Overview – Reference Page 261

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Raw Mill Fan	bearing temperature	As per OEM Manual	Daily
Raw Mill Fan	air flow	Within Design Limit	Weekly
ID Fan	pressure	Normal	Hourly
Clinker Cooler	vibration	Normal	Daily
Ball Mill	motor current	Within Design Limit	Daily

Crusher House Operations – Reference Page 262

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Cement Grinding Mill Operations – Reference Page 263

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 264

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	power consumption	Normal	Hourly
Vertical Roller Mill	vibration	Normal	Hourly
Bag Filter	pressure	Normal	Weekly
Packing Machine	air flow	Normal	Monthly
Raw Mill Fan	air flow	Normal	Weekly

Cement Grinding Mill Operations – Reference Page 265

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	air flow observed in Bucket Elevator during routine inspection.	Adjusted process parameter
02:00	motor current observed in Limestone Crusher during routine inspection.	Reported to control room engineer
04:00	motor current observed in Limestone Crusher during routine inspection.	Within normal range
06:00	pressure observed in Packing Machine during routine inspection.	Within normal range
08:00	bearing temperature observed in Cement Separator during routine inspection.	Maintenance notified
10:00	power consumption	Reported to control room

observed in Limestone Crusher during routine inspection.	engineer

Safety and Permit Procedures – Reference Page 266

The Clinker Cooler must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 267

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Packing Machine	feed rate	Normal	Daily
Vertical Roller Mill	temperature	As per OEM Manual	Monthly
Packing Machine	vibration	Normal	Daily
Clinker Cooler	temperature	Normal	Hourly
Raw Mill Fan	feed rate	Within Design Limit	Hourly

Rotary Kiln Operations Manual – Reference Page 268

The Ball Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 269

The ID Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 270

The Rotary Kiln must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Packing Machine	bearing temperature	As per OEM Manual	Weekly
Raw Mill Fan	power consumption	As per OEM Manual	Daily
Raw Mill Fan	feed rate	As per OEM Manual	Monthly
Bucket Elevator	temperature	As per OEM Manual	Hourly
Cement Separator	temperature	Within Design Limit	Daily

Shift Log Example

Time	Observation	Action Taken
00:00	bearing temperature observed in Raw Mill Fan during routine inspection.	Reported to control room engineer
02:00	motor current observed in Rotary Kiln during routine inspection.	Adjusted process parameter
04:00	motor current observed in	Adjusted process parameter

	Cement Separator during routine inspection.	
06:00	motor current observed in ID Fan during routine inspection.	Maintenance notified
08:00	bearing temperature observed in Limestone Crusher during routine inspection.	Adjusted process parameter
10:00	power consumption observed in Cement Separator during routine inspection.	Maintenance notified

Environmental Monitoring Procedures – Reference Page 271

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Troubleshooting Guide – Reference Page 272

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 273

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bag Filter	vibration	As per OEM Manual	Daily
Electrostatic Precipitator	vibration	As per OEM Manual	Daily
Clinker Cooler	temperature	As per OEM Manual	Daily
Vertical Roller Mill	vibration	As per OEM Manual	Monthly
Cement Separator	motor current	As per OEM Manual	Hourly

Equipment Maintenance Sheets – Reference Page 274

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 275

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	feed rate observed in Raw Mill Fan during routine inspection.	Reported to control room engineer
02:00	power consumption observed in Raw Mill Fan during routine inspection.	Within normal range
04:00	power consumption observed in Clinker Cooler during routine inspection.	Adjusted process parameter
06:00	temperature observed in ID Fan during routine inspection.	Adjusted process parameter
08:00	motor current observed in ID Fan during routine inspection.	Reported to control room engineer
10:00	vibration observed in Raw Mill Fan during routine	Maintenance notified

inspection.

Crusher House Operations – Reference Page 276

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	air flow	Normal	Daily
Rotary Kiln	pressure	Normal	Daily
Electrostatic Precipitator	pressure	Within Design Limit	Daily
ID Fan	pressure	Within Design Limit	Hourly
Bag Filter	bearing temperature	Within Design Limit	Weekly

Plant Overview – Reference Page 277

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 278

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 279

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	bearing temperature	Normal	Monthly
ID Fan	power consumption	Within Design Limit	Monthly
Ball Mill	vibration	Within Design Limit	Weekly
Clinker Cooler	power consumption	Within Design Limit	Weekly
Cement Separator	feed rate	Within Design Limit	Monthly

Limestone Quarry Operations Manual – Reference Page 280

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	bearing temperature observed in Bucket Elevator during routine inspection.	Within normal range
02:00	vibration observed in Rotary Kiln during routine inspection.	Maintenance notified
04:00	power consumption observed in Clinker Cooler during routine inspection.	Adjusted process parameter
06:00	power consumption observed in Limestone Crusher during routine inspection.	Reported to control room engineer
08:00	air flow observed in Electrostatic Precipitator during routine inspection.	Adjusted process parameter
10:00	vibration observed in Clinker	Maintenance notified

Cooler during routine inspection.

Shift Handover and Operations Logs – Reference Page 281

The Vertical Roller Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 282

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Vertical Roller Mill	power consumption	Normal	Monthly
Limestone Crusher	motor current	Normal	Hourly
Limestone Crusher	temperature	Within Design Limit	Monthly
Packing Machine	vibration	Within Design Limit	Hourly
Vertical Roller Mill	pressure	Normal	Weekly

Shift Handover and Operations Logs – Reference Page 283

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 284

The Packing Machine must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 285

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	power consumption	Within Design Limit	Monthly
Vertical Roller Mill	feed rate	As per OEM Manual	Hourly
Electrostatic Precipitator	air flow	As per OEM Manual	Daily
Limestone Crusher	pressure	Normal	Daily
Vertical Roller Mill	vibration	Normal	Hourly

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Bucket Elevator during routine inspection.	Adjusted process parameter
02:00	vibration observed in ID Fan during routine inspection.	Reported to control room engineer
04:00	feed rate observed in	Adjusted process parameter

	Electrostatic Precipitator during routine inspection.	
06:00	bearing temperature observed in Vertical Roller Mill during routine inspection.	Maintenance notified
08:00	feed rate observed in Bucket Elevator during routine inspection.	Maintenance notified
10:00	feed rate observed in Vertical Roller Mill during routine inspection.	Adjusted process parameter

Clinker Cooler Operations – Reference Page 286

The Vertical Roller Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 287

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 288

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bag Filter	air flow	Within Design Limit	Weekly
Clinker Cooler	temperature	Normal	Daily
Rotary Kiln	temperature	Normal	Monthly
Cement Separator	bearing temperature	Within Design Limit	Hourly
Vertical Roller Mill	power consumption	Within Design Limit	Monthly

Quality Control Laboratory SOP – Reference Page 289

The Ball Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 290

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Bag Filter during routine inspection.	Adjusted process parameter
02:00	temperature observed in Ball Mill during routine inspection.	Maintenance notified
04:00	motor current observed in ID Fan during routine inspection.	Within normal range
06:00	motor current observed in Rotary Kiln during routine inspection.	Reported to control room engineer
08:00	power consumption observed in Bucket Elevator during routine inspection.	Within normal range
10:00	bearing temperature observed in Clinker Cooler	Reported to control room engineer

during routine inspection.

Safety and Permit Procedures – Reference Page 291

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	motor current	Normal	Hourly
Ball Mill	vibration	Within Design Limit	Hourly
Limestone Crusher	feed rate	Normal	Weekly
Limestone Crusher	air flow	Within Design Limit	Hourly
Vertical Roller Mill	temperature	As per OEM Manual	Daily

Plant Overview – Reference Page 292

The Rotary Kiln must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 293

The Vertical Roller Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 294

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Electrostatic Precipitator	power consumption	Within Design Limit	Daily
Packing Machine	power consumption	Within Design Limit	Monthly
ID Fan	power consumption	Normal	Monthly
Electrostatic Precipitator	bearing temperature	As per OEM Manual	Weekly
Ball Mill	air flow	As per OEM Manual	Weekly

Plant Overview – Reference Page 295

The ID Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	power consumption observed in Vertical Roller Mill during routine inspection.	Adjusted process parameter
02:00	temperature observed in ID Fan during routine inspection.	Adjusted process parameter
04:00	air flow observed in Cement Separator during routine inspection.	Maintenance notified
06:00	pressure observed in Clinker Cooler during routine inspection.	Reported to control room engineer
08:00	pressure observed in Bucket Elevator during routine inspection.	Maintenance notified
10:00	temperature observed in Ball	Adjusted process parameter

Mill during routine
inspection.

Raw Mill Operations (VRM) – Reference Page 296

The Vertical Roller Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preventive Maintenance Manual – Reference Page 297

The Vertical Roller Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bag Filter	air flow	As per OEM Manual	Weekly
Limestone Crusher	pressure	As per OEM Manual	Hourly
Limestone Crusher	bearing temperature	Normal	Hourly
Bucket Elevator	pressure	As per OEM Manual	Weekly
Ball Mill	pressure	As per OEM Manual	Daily

Packing Plant Operations – Reference Page 298

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 299

The Rotary Kiln must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 300

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Packing Machine	feed rate	Normal	Monthly
Bucket Elevator	temperature	Normal	Hourly
Clinker Cooler	motor current	Normal	Weekly
ID Fan	air flow	Normal	Monthly
Ball Mill	bearing temperature	Normal	Daily

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Cement Separator during routine inspection.	Reported to control room engineer
02:00	air flow observed in Ball Mill during routine inspection.	Maintenance notified
04:00	temperature observed in Rotary Kiln during routine	Maintenance notified

	inspection.	
06:00	temperature observed in Rotary Kiln during routine inspection.	Reported to control room engineer
08:00	motor current observed in Electrostatic Precipitator during routine inspection.	Adjusted process parameter
10:00	vibration observed in Bucket Elevator during routine inspection.	Adjusted process parameter

Preventive Maintenance Manual – Reference Page 301

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 302

The Electrostatic Precipitator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 303

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Packing Machine	pressure	As per OEM Manual	Monthly
Cement Separator	feed rate	Normal	Weekly
Packing Machine	bearing temperature	Within Design Limit	Daily
Electrostatic Precipitator	vibration	Within Design Limit	Monthly
Clinker Cooler	power consumption	Within Design Limit	Daily

Safety and Permit Procedures – Reference Page 304

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 305

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Bucket Elevator during routine inspection.	Adjusted process parameter
02:00	motor current observed in Electrostatic Precipitator during routine inspection.	Within normal range
04:00	air flow observed in Electrostatic Precipitator during routine inspection.	Maintenance notified
06:00	temperature observed in Vertical Roller Mill during routine inspection.	Maintenance notified
08:00	pressure observed in Clinker Cooler during routine inspection.	Reported to control room engineer
10:00	air flow observed in Rotary Kiln during routine	Maintenance notified

inspection.

Rotary Kiln Operations Manual – Reference Page 306

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	air flow	Normal	Daily
Bag Filter	power consumption	Within Design Limit	Daily
Rotary Kiln	bearing temperature	As per OEM Manual	Monthly
Bag Filter	feed rate	Normal	Daily
Raw Mill Fan	power consumption	Normal	Weekly

Clinker Cooler Operations – Reference Page 307

The Bag Filter must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Cement Grinding Mill Operations – Reference Page 308

The Vertical Roller Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Handover and Operations Logs – Reference Page 309

The Clinker Cooler must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
ID Fan	pressure	As per OEM Manual	Daily
Bucket Elevator	pressure	Normal	Monthly
Cement Separator	temperature	Normal	Hourly
Packing Machine	pressure	As per OEM Manual	Weekly
Ball Mill	feed rate	Normal	Weekly

Equipment Maintenance Sheets – Reference Page 310

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Rotary Kiln during routine inspection.	Maintenance notified
02:00	temperature observed in Vertical Roller Mill during routine inspection.	Within normal range
04:00	power consumption observed in Cement Separator during routine inspection.	Adjusted process parameter
06:00	vibration observed in Clinker Cooler during routine inspection.	Reported to control room engineer
08:00	power consumption observed in Bag Filter during routine inspection.	Maintenance notified
10:00	power consumption	Adjusted process parameter

observed in Limestone
Crusher during routine
inspection.

Clinker Cooler Operations – Reference Page 311

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 312

The Ball Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Vertical Roller Mill	power consumption	As per OEM Manual	Hourly
Limestone Crusher	feed rate	As per OEM Manual	Weekly
ID Fan	feed rate	As per OEM Manual	Hourly
Electrostatic Precipitator	motor current	As per OEM Manual	Daily
Limestone Crusher	vibration	Within Design Limit	Weekly

Limestone Quarry Operations Manual – Reference Page 313

The Packing Machine must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Packing Plant Operations – Reference Page 314

The Raw Mill Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Handover and Operations Logs – Reference Page 315

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	air flow	Normal	Daily
Limestone Crusher	pressure	Normal	Weekly
Vertical Roller Mill	vibration	Within Design Limit	Daily
Bucket Elevator	bearing temperature	Normal	Daily
Rotary Kiln	air flow	As per OEM Manual	Weekly

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Ball Mill during routine inspection.	Maintenance notified
02:00	air flow observed in Raw Mill Fan during routine inspection.	Adjusted process parameter
04:00	feed rate observed in Vertical Roller Mill during routine	Reported to control room engineer

	inspection.	
06:00	motor current observed in Rotary Kiln during routine inspection.	Within normal range
08:00	temperature observed in Limestone Crusher during routine inspection.	Within normal range
10:00	vibration observed in Bag Filter during routine inspection.	Within normal range

Crusher House Operations – Reference Page 316

The Bag Filter must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 317

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 318

The Rotary Kiln must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Cement Separator	vibration	Within Design Limit	Daily
Bag Filter	feed rate	As per OEM Manual	Daily
Raw Mill Fan	feed rate	Normal	Daily
Vertical Roller Mill	temperature	Within Design Limit	Weekly
Limestone Crusher	temperature	Within Design Limit	Hourly

Rotary Kiln Operations Manual – Reference Page 319

The ID Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 320

The Limestone Crusher must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	power consumption observed in Vertical Roller Mill during routine inspection.	Within normal range
02:00	feed rate observed in Clinker Cooler during routine inspection.	Maintenance notified
04:00	temperature observed in Packing Machine during routine inspection.	Within normal range
06:00	motor current observed in Bag Filter during routine inspection.	Reported to control room engineer
08:00	power consumption observed in Ball Mill during routine inspection.	Reported to control room engineer
10:00	pressure observed in	Maintenance notified

Limestone Crusher during
routine inspection.

Preheater and Calciner System – Reference Page 321

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Rotary Kiln	vibration	Within Design Limit	Daily
Bag Filter	temperature	As per OEM Manual	Weekly
ID Fan	pressure	Normal	Weekly
Cement Separator	air flow	Within Design Limit	Hourly
Clinker Cooler	power consumption	As per OEM Manual	Monthly

Limestone Quarry Operations Manual – Reference Page 322

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Environmental Monitoring Procedures – Reference Page 323

The Vertical Roller Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 324

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Raw Mill Fan	vibration	Within Design Limit	Weekly
Raw Mill Fan	temperature	As per OEM Manual	Weekly
Clinker Cooler	bearing temperature	Normal	Monthly
Packing Machine	motor current	Normal	Weekly
Rotary Kiln	power consumption	Normal	Weekly

Preheater and Calciner System – Reference Page 325

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Electrostatic Precipitator during routine inspection.	Maintenance notified
02:00	air flow observed in Bucket Elevator during routine inspection.	Reported to control room engineer
04:00	motor current observed in Bag Filter during routine inspection.	Reported to control room engineer
06:00	temperature observed in Cement Separator during routine inspection.	Adjusted process parameter
08:00	feed rate observed in ID Fan during routine inspection.	Within normal range
10:00	pressure observed in Raw Mill Fan during routine inspection.	Maintenance notified

Packing Plant Operations – Reference Page 326

The Clinker Cooler must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 327

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
ID Fan	motor current	Within Design Limit	Daily
Raw Mill Fan	air flow	Within Design Limit	Hourly
Limestone Crusher	air flow	As per OEM Manual	Hourly
Clinker Cooler	pressure	Normal	Daily
Limestone Crusher	power consumption	Normal	Daily

Equipment Maintenance Sheets – Reference Page 328

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Plant Overview – Reference Page 329

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Cement Grinding Mill Operations – Reference Page 330

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	feed rate	Normal	Hourly
Vertical Roller Mill	motor current	Normal	Hourly
Clinker Cooler	pressure	Normal	Weekly
Electrostatic Precipitator	bearing temperature	Within Design Limit	Monthly
Bucket Elevator	vibration	As per OEM Manual	Hourly

Shift Log Example

Time	Observation	Action Taken
00:00	motor current observed in Limestone Crusher during routine inspection.	Adjusted process parameter
02:00	pressure observed in Packing Machine during routine inspection.	Adjusted process parameter

04:00	pressure observed in ID Fan during routine inspection.	Adjusted process parameter
06:00	vibration observed in Ball Mill during routine inspection.	Maintenance notified
08:00	vibration observed in Rotary Kiln during routine inspection.	Maintenance notified
10:00	air flow observed in Packing Machine during routine inspection.	Maintenance notified

Cement Grinding Mill Operations – Reference Page 331

The Bucket Elevator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 332

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Cement Grinding Mill Operations – Reference Page 333

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
ID Fan	feed rate	As per OEM Manual	Weekly
Vertical Roller Mill	bearing temperature	Within Design Limit	Weekly
Raw Mill Fan	power consumption	As per OEM Manual	Daily
Vertical Roller Mill	vibration	As per OEM Manual	Hourly
Ball Mill	bearing temperature	Within Design Limit	Monthly

Troubleshooting Guide – Reference Page 334

The Bucket Elevator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 335

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Cement Separator during routine inspection.	Maintenance notified
02:00	temperature observed in Electrostatic Precipitator during routine inspection.	Within normal range
04:00	vibration observed in Electrostatic Precipitator during routine inspection.	Maintenance notified
06:00	vibration observed in Clinker Cooler during routine inspection.	Adjusted process parameter
08:00	bearing temperature observed in Ball Mill during routine inspection.	Reported to control room engineer
10:00	temperature observed in Cement Separator during	Adjusted process parameter

routine inspection.

Limestone Quarry Operations Manual – Reference Page 336

The Cement Separator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	motor current	Within Design Limit	Monthly
Rotary Kiln	temperature	Normal	Monthly
Bucket Elevator	bearing temperature	Within Design Limit	Daily
Ball Mill	power consumption	Normal	Monthly
Cement Separator	motor current	As per OEM Manual	Monthly

Troubleshooting Guide – Reference Page 337

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 338

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Plant Overview – Reference Page 339

The Electrostatic Precipitator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Raw Mill Fan	vibration	Normal	Daily
Bucket Elevator	temperature	As per OEM Manual	Monthly
Clinker Cooler	power consumption	Within Design Limit	Monthly
Electrostatic Precipitator	air flow	Normal	Hourly
Cement Separator	bearing temperature	As per OEM Manual	Monthly

Shift Handover and Operations Logs – Reference Page 340

The Rotary Kiln must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	bearing temperature observed in Limestone Crusher during routine inspection.	Adjusted process parameter
02:00	pressure observed in Rotary Kiln during routine inspection.	Reported to control room engineer
04:00	vibration observed in Cement Separator during routine inspection.	Reported to control room engineer
06:00	feed rate observed in Bucket Elevator during routine inspection.	Maintenance notified
08:00	temperature observed in Electrostatic Precipitator during routine inspection.	Reported to control room engineer
10:00	vibration observed in Cement	Maintenance notified

Separator during routine
inspection.

Packing Plant Operations – Reference Page 341

The Ball Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Troubleshooting Guide – Reference Page 342

The Packing Machine must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	feed rate	As per OEM Manual	Monthly
ID Fan	bearing temperature	Normal	Monthly
Bucket Elevator	air flow	Normal	Monthly
Raw Mill Fan	vibration	Within Design Limit	Hourly
Limestone Crusher	power consumption	Within Design Limit	Hourly

Quality Control Laboratory SOP – Reference Page 343

The Rotary Kiln must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 344

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 345

The Bucket Elevator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	motor current	As per OEM Manual	Daily
Electrostatic Precipitator	air flow	As per OEM Manual	Weekly
ID Fan	bearing temperature	Normal	Weekly
Rotary Kiln	feed rate	As per OEM Manual	Hourly
Rotary Kiln	motor current	As per OEM Manual	Monthly

Shift Log Example

Time	Observation	Action Taken
00:00	pressure observed in ID Fan during routine inspection.	Adjusted process parameter
02:00	feed rate observed in ID Fan during routine inspection.	Adjusted process parameter
04:00	feed rate observed in Vertical Roller Mill during routine	Within normal range

	inspection.	
06:00	motor current observed in Rotary Kiln during routine inspection.	Within normal range
08:00	pressure observed in Ball Mill during routine inspection.	Adjusted process parameter
10:00	power consumption observed in Ball Mill during routine inspection.	Reported to control room engineer

Equipment Maintenance Sheets – Reference Page 346

The Bucket Elevator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 347

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 348

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	temperature	Normal	Hourly
Bag Filter	vibration	Within Design Limit	Monthly
Limestone Crusher	temperature	As per OEM Manual	Weekly
Limestone Crusher	air flow	As per OEM Manual	Monthly
Electrostatic Precipitator	vibration	Within Design Limit	Monthly

Troubleshooting Guide – Reference Page 349

The Clinker Cooler must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 350

The Bucket Elevator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	bearing temperature observed in Vertical Roller Mill during routine inspection.	Adjusted process parameter
02:00	air flow observed in ID Fan during routine inspection.	Adjusted process parameter
04:00	vibration observed in Packing Machine during routine inspection.	Adjusted process parameter
06:00	temperature observed in Packing Machine during routine inspection.	Reported to control room engineer
08:00	feed rate observed in Limestone Crusher during routine inspection.	Maintenance notified
10:00	air flow observed in Cement Separator during routine	Reported to control room engineer

inspection.

Troubleshooting Guide – Reference Page 351

The Vertical Roller Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Packing Machine	pressure	Within Design Limit	Daily
Clinker Cooler	pressure	As per OEM Manual	Weekly
Ball Mill	vibration	Normal	Monthly
Rotary Kiln	pressure	Within Design Limit	Weekly
Ball Mill	bearing temperature	As per OEM Manual	Monthly

Quality Control Laboratory SOP – Reference Page 352

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Environmental Monitoring Procedures – Reference Page 353

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 354

The Limestone Crusher must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	bearing temperature	As per OEM Manual	Daily
Packing Machine	temperature	Within Design Limit	Hourly
Bucket Elevator	vibration	Within Design Limit	Weekly
Cement Separator	vibration	Within Design Limit	Hourly
Limestone Crusher	vibration	Normal	Daily

Environmental Monitoring Procedures – Reference Page 355

The ID Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	power consumption observed in Electrostatic Precipitator during routine inspection.	Adjusted process parameter
02:00	pressure observed in Electrostatic Precipitator during routine inspection.	Reported to control room engineer
04:00	temperature observed in Limestone Crusher during routine inspection.	Within normal range
06:00	bearing temperature observed in Bucket Elevator during routine inspection.	Maintenance notified
08:00	feed rate observed in ID Fan during routine inspection.	Within normal range
10:00	vibration observed in Electrostatic Precipitator	Reported to control room engineer

during routine inspection.

Environmental Monitoring Procedures – Reference Page 356

The Packing Machine must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Clinker Cooler Operations – Reference Page 357

The Cement Separator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	air flow	Normal	Daily
Clinker Cooler	power consumption	Normal	Monthly
Raw Mill Fan	bearing temperature	As per OEM Manual	Daily
Bucket Elevator	motor current	Within Design Limit	Hourly
Clinker Cooler	bearing temperature	As per OEM Manual	Weekly

Environmental Monitoring Procedures – Reference Page 358

The Limestone Crusher must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 359

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 360

The Ball Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
ID Fan	bearing temperature	Normal	Monthly
Bucket Elevator	feed rate	Normal	Weekly
ID Fan	motor current	As per OEM Manual	Monthly
Bag Filter	temperature	Within Design Limit	Hourly
Cement Separator	motor current	As per OEM Manual	Hourly

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Rotary Kiln during routine inspection.	Reported to control room engineer
02:00	pressure observed in Raw Mill Fan during routine inspection.	Within normal range
04:00	vibration observed in Clinker	Maintenance notified

	Cooler during routine inspection.	
06:00	temperature observed in Electrostatic Precipitator during routine inspection.	Adjusted process parameter
08:00	air flow observed in Packing Machine during routine inspection.	Reported to control room engineer
10:00	feed rate observed in Vertical Roller Mill during routine inspection.	Reported to control room engineer

Raw Mill Operations (VRM) – Reference Page 361

The Bucket Elevator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Plant Overview – Reference Page 362

The Limestone Crusher must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 363

The Clinker Cooler must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Clinker Cooler	motor current	Normal	Daily
Electrostatic Precipitator	vibration	Normal	Hourly
Bag Filter	motor current	Within Design Limit	Weekly
Bag Filter	vibration	Normal	Monthly
Cement Separator	pressure	Normal	Monthly

Shift Handover and Operations Logs – Reference Page 364

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Limestone Quarry Operations Manual – Reference Page 365

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	air flow observed in Ball Mill during routine inspection.	Adjusted process parameter
02:00	temperature observed in Clinker Cooler during routine inspection.	Adjusted process parameter
04:00	motor current observed in Vertical Roller Mill during routine inspection.	Within normal range
06:00	motor current observed in ID Fan during routine inspection.	Maintenance notified
08:00	bearing temperature observed in Electrostatic Precipitator during routine inspection.	Maintenance notified
10:00	air flow observed in Cement Separator during routine	Reported to control room engineer

inspection.

Cement Grinding Mill Operations – Reference Page 366

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Cement Separator	bearing temperature	Within Design Limit	Weekly
Ball Mill	power consumption	Within Design Limit	Hourly
Rotary Kiln	vibration	As per OEM Manual	Daily
Vertical Roller Mill	vibration	As per OEM Manual	Daily
Clinker Cooler	bearing temperature	Within Design Limit	Hourly

Preheater and Calciner System – Reference Page 367

The Vertical Roller Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Raw Mill Operations (VRM) – Reference Page 368

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 369

The Bag Filter must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Cement Separator	air flow	Within Design Limit	Hourly
Bag Filter	motor current	Within Design Limit	Weekly
Ball Mill	feed rate	As per OEM Manual	Daily
Clinker Cooler	temperature	Normal	Monthly
Rotary Kiln	bearing temperature	As per OEM Manual	Monthly

Rotary Kiln Operations Manual – Reference Page 370

The Bag Filter must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	air flow observed in Cement Separator during routine inspection.	Reported to control room engineer
02:00	bearing temperature observed in Vertical Roller Mill during routine inspection.	Adjusted process parameter
04:00	vibration observed in Clinker Cooler during routine inspection.	Adjusted process parameter
06:00	temperature observed in ID Fan during routine inspection.	Maintenance notified
08:00	bearing temperature observed in Raw Mill Fan during routine inspection.	Adjusted process parameter
10:00	feed rate observed in Raw	Within normal range

Mill Fan during routine inspection.

Raw Mill Operations (VRM) – Reference Page 371

The Bucket Elevator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 372

The Packing Machine must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Vertical Roller Mill	bearing temperature	Normal	Hourly
Raw Mill Fan	bearing temperature	Normal	Daily
Raw Mill Fan	power consumption	Within Design Limit	Daily
Cement Separator	motor current	As per OEM Manual	Daily
ID Fan	vibration	Within Design Limit	Monthly

Troubleshooting Guide – Reference Page 373

The ID Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Safety and Permit Procedures – Reference Page 374

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Troubleshooting Guide – Reference Page 375

The Vertical Roller Mill must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Rotary Kiln	power consumption	Normal	Hourly
Clinker Cooler	air flow	Normal	Weekly
Ball Mill	vibration	Normal	Hourly
Limestone Crusher	bearing temperature	As per OEM Manual	Daily
Vertical Roller Mill	air flow	Within Design Limit	Hourly

Shift Log Example

Time	Observation	Action Taken
00:00	feed rate observed in Packing Machine during routine inspection.	Reported to control room engineer
02:00	feed rate observed in Limestone Crusher during routine inspection.	Maintenance notified
04:00	power consumption	Maintenance notified

	observed in ID Fan during routine inspection.	
06:00	air flow observed in Rotary Kiln during routine inspection.	Reported to control room engineer
08:00	vibration observed in Bag Filter during routine inspection.	Reported to control room engineer
10:00	pressure observed in Packing Machine during routine inspection.	Maintenance notified

Clinker Cooler Operations – Reference Page 376

The Ball Mill must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 377

The Bag Filter must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preventive Maintenance Manual – Reference Page 378

The Bag Filter must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bucket Elevator	feed rate	Within Design Limit	Monthly
Rotary Kiln	pressure	Normal	Hourly
Packing Machine	bearing temperature	Within Design Limit	Daily
Vertical Roller Mill	pressure	Normal	Daily
Ball Mill	feed rate	As per OEM Manual	Monthly

Preventive Maintenance Manual – Reference Page 379

The Clinker Cooler must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Cement Grinding Mill Operations – Reference Page 380

The ID Fan must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	power consumption observed in Raw Mill Fan during routine inspection.	Within normal range
02:00	temperature observed in Rotary Kiln during routine inspection.	Within normal range
04:00	motor current observed in Packing Machine during routine inspection.	Within normal range
06:00	temperature observed in Raw Mill Fan during routine inspection.	Maintenance notified
08:00	vibration observed in Ball Mill during routine inspection.	Reported to control room engineer
10:00	motor current observed in Bag Filter during routine inspection.	Maintenance notified

Plant Overview – Reference Page 381

The Ball Mill must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Cement Separator	temperature	As per OEM Manual	Daily
Bucket Elevator	feed rate	Normal	Hourly
Packing Machine	motor current	Normal	Hourly
Bucket Elevator	vibration	Within Design Limit	Hourly
Bucket Elevator	pressure	Within Design Limit	Monthly

Troubleshooting Guide – Reference Page 382

The Cement Separator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Environmental Monitoring Procedures – Reference Page 383

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 384

The Electrostatic Precipitator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Packing Machine	motor current	Normal	Hourly
Packing Machine	pressure	Normal	Hourly
Vertical Roller Mill	temperature	Normal	Monthly
Packing Machine	pressure	As per OEM Manual	Monthly
Ball Mill	air flow	Normal	Hourly

Clinker Cooler Operations – Reference Page 385

The Vertical Roller Mill must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	temperature observed in Bucket Elevator during routine inspection.	Adjusted process parameter
02:00	air flow observed in Limestone Crusher during routine inspection.	Within normal range
04:00	temperature observed in Clinker Cooler during routine inspection.	Adjusted process parameter
06:00	vibration observed in Bag Filter during routine inspection.	Maintenance notified
08:00	vibration observed in Raw Mill Fan during routine inspection.	Reported to control room engineer
10:00	power consumption observed in ID Fan during	Within normal range

routine inspection.

Crusher House Operations – Reference Page 386

The Limestone Crusher must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Quality Control Laboratory SOP – Reference Page 387

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Ball Mill	feed rate	Within Design Limit	Hourly
Raw Mill Fan	vibration	Normal	Monthly
Cement Separator	feed rate	Normal	Weekly
Ball Mill	vibration	As per OEM Manual	Weekly
Electrostatic Precipitator	vibration	Normal	Monthly

Raw Mill Operations (VRM) – Reference Page 388

The Bucket Elevator must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Vertical Roller Mill must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Crusher House Operations – Reference Page 389

The Electrostatic Precipitator must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Clinker Cooler must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Clinker Cooler Operations – Reference Page 390

The Clinker Cooler must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Bag Filter	vibration	As per OEM Manual	Weekly
Limestone Crusher	air flow	Normal	Hourly
Vertical Roller Mill	motor current	As per OEM Manual	Weekly
Bag Filter	temperature	Normal	Weekly
Rotary Kiln	vibration	As per OEM Manual	Monthly

Shift Log Example

Time	Observation	Action Taken
00:00	vibration observed in Ball Mill during routine inspection.	Reported to control room engineer
02:00	motor current observed in ID Fan during routine inspection.	Maintenance notified
04:00	motor current observed in Packing Machine during	Adjusted process parameter

	routine inspection.	
06:00	feed rate observed in Electrostatic Precipitator during routine inspection.	Reported to control room engineer
08:00	pressure observed in Ball Mill during routine inspection.	Reported to control room engineer
10:00	vibration observed in Vertical Roller Mill during routine inspection.	Reported to control room engineer

Crusher House Operations – Reference Page 391

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Raw Mill Fan must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 392

The Electrostatic Precipitator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Equipment Maintenance Sheets – Reference Page 393

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Limestone Crusher	pressure	Normal	Hourly
Cement Separator	air flow	Within Design Limit	Daily
Packing Machine	air flow	As per OEM Manual	Weekly
Cement Separator	power consumption	Normal	Hourly
Clinker Cooler	vibration	Within Design Limit	Daily

Shift Handover and Operations Logs – Reference Page 394

The Vertical Roller Mill must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Ball Mill must be monitored continuously for temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Preheater and Calciner System – Reference Page 395

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for air flow stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in air flow must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	air flow observed in Cement Separator during routine inspection.	Within normal range
02:00	feed rate observed in Raw Mill Fan during routine inspection.	Reported to control room engineer
04:00	feed rate observed in Bucket Elevator during routine inspection.	Reported to control room engineer
06:00	power consumption observed in Vertical Roller Mill during routine inspection.	Maintenance notified
08:00	pressure observed in Bucket Elevator during routine inspection.	Within normal range
10:00	vibration observed in Bag	Reported to control room

Filter during routine inspection.	engineer

Shift Handover and Operations Logs – Reference Page 396

The Packing Machine must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Electrostatic Precipitator must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Cement Separator	bearing temperature	Within Design Limit	Daily
Ball Mill	power consumption	Normal	Weekly
Cement Separator	pressure	Normal	Weekly
ID Fan	power consumption	As per OEM Manual	Daily
Bag Filter	feed rate	Within Design Limit	Monthly

Troubleshooting Guide – Reference Page 397

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bucket Elevator must be monitored continuously for power consumption stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in power consumption must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Packing Machine must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Rotary Kiln Operations Manual – Reference Page 398

The Packing Machine must be monitored continuously for pressure stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in pressure must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Limestone Crusher must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Cement Separator must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Handover and Operations Logs – Reference Page 399

The Limestone Crusher must be monitored continuously for motor current stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in motor current must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Rotary Kiln must be monitored continuously for vibration stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in vibration must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The Bag Filter must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Operational Monitoring Table

Equipment	Parameter	Normal Range	Inspection Frequency
Raw Mill Fan	vibration	Within Design Limit	Daily
Packing Machine	feed rate	Within Design Limit	Daily
Clinker Cooler	vibration	As per OEM Manual	Weekly
Bag Filter	bearing temperature	As per OEM Manual	Weekly
Clinker Cooler	air flow	As per OEM Manual	Monthly

Safety and Permit Procedures – Reference Page 400

The Cement Separator must be monitored continuously for bearing temperature stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in bearing temperature must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

The ID Fan must be monitored continuously for feed rate stability during plant operation. Operators should record readings in the distributed control system and verify that values remain within design limits. Any abnormal change in feed rate must be reported to the control room and logged in the shift report. Preventive inspection helps ensure stable clinker production and consistent cement quality.

Shift Log Example

Time	Observation	Action Taken
00:00	power consumption observed in Ball Mill during routine inspection.	Within normal range
02:00	feed rate observed in Electrostatic Precipitator during routine inspection.	Within normal range
04:00	power consumption observed in Electrostatic Precipitator during routine inspection.	Adjusted process parameter
06:00	feed rate observed in Bucket Elevator during routine inspection.	Maintenance notified
08:00	motor current observed in Ball Mill during routine inspection.	Maintenance notified
10:00	feed rate observed in Vertical	Maintenance notified

Roller Mill during routine inspection.